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Exercise 2.2

1) Find the norm $\|P\|$ of

$$P = \{-1, 0, 1, 3, 4, 4, 1, 5\}, [a, b] = [-1, 5]$$

$$0 - (-1) = 1$$

$$1 \cdot 3 - 0 = 1 \cdot 3$$

$$4 - 1 \cdot 3 = 2 \cdot 7$$

$$~~4~~ 4 \cdot 1 - 4 = 0 \cdot 1$$

$$5 - 4 \cdot 1 = 0 \cdot 9$$

$$\|P\| = \max \{1, 1 \cdot 3, 2 \cdot 7, 0 \cdot 1, 0 \cdot 9\}$$

$$= 2 \cdot 7$$

3) Find the norm $\|P\|$ of

$$P = \{-3, 0, 2 \cdot 3, 4 \cdot 6, 4 \cdot 8, 5 \cdot 5, 6\}$$

$$[a, b] = [-3, 6]$$

$$2 \cdot 3 - 0 = 0 - (-3) = 3$$

$$~~3 \cdot 5 - 3 = 0 \cdot 5~~ 2 \cdot 3$$

$$4 \cdot 6 - ~~3 \cdot 5~~ = ~~0 \cdot 5~~ 2 \cdot 3$$

$$4 \cdot 8 - 4 \cdot 6 = 0 \cdot 2$$

$$~~4 \cdot 8 - 4 = 0 \cdot 9~~$$

$$5 \cdot 5 - 4 \cdot 8 = 0 \cdot 7$$

$$6 - 5 \cdot 5 = 0 \cdot 5$$

$$\|P\| = 3$$

5) Find the norm $\|P\|$ of

$$P = \{3, 3.5, 3.6, 4, 4.9, 7\},$$

$$[a, b] = [3, 7]$$

$$3.5 - 3 = 0.5$$

$$3.6 - 3.5 = 0.1$$

$$4 - 3.6 = 0.4$$

$$4.9 - 4 = 0.9$$

$$7 - 4.9 = 2.1$$

$$\|P\| = 2.1$$

7) Find the norm $\|P\|$ of

$$P = \{-1, -\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2}, 2\},$$

$$[a, b] = [-1, 2]$$

$$x_i - x_{i-1} = \frac{1}{2} \text{ for all subintervals}$$

$$\|P\| = \frac{1}{2}$$

10) Define a regular partition $[-1, 4]$ into $n = 6$ subintervals

$$\Delta x = \frac{4 - (-1)}{6} = \frac{5}{6}$$

$$x_i = -1 + i \cdot \frac{5}{6},$$

$$i = 0, \dots, 6$$

$$P = \{-1, -\frac{1}{6}, \frac{2}{3}, \frac{3}{2}, \frac{7}{3}, \frac{19}{6}, 4\}$$

12) Define a regular partition
of $[0, 1]$ into $n = 4$
 $\Delta x = \frac{1}{4}$

$$P = \left\{ 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1 \right\}$$

~~13~~ Find the Riemann sum using
Interval $[0, 4]$
13) Left endpoints

$$f(x) = x^2 + 1, \quad P = \{0, 1, 3, 4\}$$

widths 1, 2, 1

$$\begin{aligned} R &= f(0)(1) + f(1)(2) + f(3)(1) \\ &= 1 + 4 + 10 \\ &= 15 \end{aligned}$$

14) Using right endpoints

$$\begin{aligned} R &= f(1)(1) + f(3)(2) + f(4)(1) \\ &= 2 + 20 + 17 \\ &= 39 \end{aligned}$$

15) Using midpoints

(midpoints $- 0.5, 2, 3.5$)

$$\begin{aligned} R &= f(0.5)(1) + f(2)(2) + f(3.5)(1) \\ &= 1.25 + 10 + 13.25 \\ &= 24.5 \end{aligned}$$

16) Find the area under

$$f(x) = \frac{x}{3} \text{ on } [1, 2]$$

$$A = \int_1^2 \frac{x}{3} dx = \frac{1}{3} \int_1^2 x dx$$

$$\frac{1}{3} \left[\frac{x^2}{2} \right]_1^2 = \frac{1}{3} \left(2 - \frac{1}{2} \right) = \frac{1}{2}$$

$$= \frac{1}{2}$$

17) Find the area under

$$f(x) = x - 1 \text{ on } [0, 3]$$

$$A = \int_0^3 (x - 1) dx = \left[\frac{x^2}{2} - x \right]_0^3$$

$$\left(\frac{9}{2} - 3 \right) = \frac{3}{2}$$

$$= \frac{3}{2}$$
